

Topic Covered:

- Probability and Counting Rules

1. A multiple-choice test contains 10 questions. There are four possible answers for each question.
 - a) In how many ways can a student answer the questions on the test if the student answers every question?
 - b) In how many ways can a student answer the questions on the test if the student can leave answers blank?

Answer:

- a) The product rule applies, since the student will perform each of 10 tasks, one after the other. There are 4 ways to do each task. Therefore, there are $4 \cdot 4 \cdot \dots \cdot 4 = 4^{10} = 1,048,576$ ways to answer the questions on the test.
- b) This is identical to part (a), except that now there are 5 ways to answer each question-give any of the 4 answers or give no answer at all. Therefore, there are $5^{10} = 9,765,625$ ways to answer the questions on the test.

2. How many positive integers are between 50 and 100:
 - a) are divisible by 7? Which integers are these?
 - b) are divisible by 11? Which integers are these?
 - c) are divisible by both 7 and 11? Which integers are these?

Answer:

- a) There are $\left\lfloor \frac{R}{d} \right\rfloor - \left\lfloor \frac{L-1}{d} \right\rfloor = \left\lfloor \frac{100}{7} \right\rfloor - \left\lfloor \frac{50-1}{7} \right\rfloor = 14 - 7 = 7$ numbers between 50 and 100 that are divisible by 7. They are 56, 63, 70, 77, 84, 91, and 98.
- b) There are $\left\lfloor \frac{100}{11} \right\rfloor - \left\lfloor \frac{50-1}{11} \right\rfloor = 9 - 4 = 5$ numbers between 50 and 100 that are divisible by 11. They are 55, 66, 77, 88, and 99.
- c) A number is divisible by both 7 and 11 if and only if it is divisible by their least common multiple, which is 77. Obviously, there is only one such number between 50 and 100, namely 77. We could also work this out as we did in the previous parts: are $\left\lfloor \frac{100}{77} \right\rfloor - \left\lfloor \frac{50-1}{77} \right\rfloor = 1 - 0 = 1$. Note also that the intersection of the sets we found in the previous two parts is precisely what we are looking for here.

3. How many one-to-one functions are there from a set with five elements to sets with the following number of elements?
 - a) 4
 - b) 5
 - c) 6
 - d) 7

Answer:

To specify such a function, we need to make 5 choices, in succession, namely the values of the function at each of the 5 elements in its domain. Therefore, the product rule applies. The first choice can be made in k ways, since any element of the codomain can be the image of the first element of the domain. After that choice has been made, there are only $k - 1$ elements of the codomain available to be the image of the second element of the domain, and so on, we see that there are $k(k - 1)(k - 2)(k - 3)(k - 4)$ one-to-one



functions from a set with 5 elements to a set with k elements.

- a) $4 \cdot 3 \cdot 2 \cdot 1 \cdot 0 = 0$.
- b) $5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$.
- c) $6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 = 720$.
- d) $7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 = 2520$.

4. In how many ways can a set of five letters be selected from the English alphabet?

Answer:

We assume that a combination is called for, not a permutation, since we are told to *select a set*, not *form an arrangement*. We need to choose 5 things from 26, so there are $C(26,5) = \binom{26}{5} = \frac{26!}{(26-5)! \cdot 5!} = \frac{26 \cdot 25 \cdot 24 \cdot 23 \cdot 22}{5!} = 65,780$ ways to do so.

5. A club has 25 members.

- a) How many ways are there to choose four members of the club to serve on an executive committee?
- b) How many ways are there to choose a president, vice president, secretary, and treasurer of the club, where no person can hold more than one office?

Answer:

- a) Since the order of choosing the members is not relevant (the offices are not differentiated), we need to use a combination. The answer is clearly $C(25,4) = \frac{25!}{(25-4)! \cdot 4!} = 12,650$.
- b) In contrast, here we need a permutation, since the order matters (we choose first a president, then a vice president, then a secretary, then a treasurer). The answer is clearly $P(25,4) = \frac{25!}{(25-4)!} = 303,600$.

6. Winning Tickets: If 50 tickets are sold and 2 prizes are to be awarded, find the probability that one person will win 2 prizes if that person buys 2 tickets.

Answer:

The probability is $\frac{2}{50} \cdot \frac{1}{49} = \frac{2}{2450} = \frac{1}{1225}$.

7. Rolling Three Dice: When 3 dice are rolled, find the probability of getting a sum of 7.

Answer:

The total number of possible outcomes when rolling 3 fair six-sided dice is $6^3 = 216$.

The favorable outcomes are the ordered triples (a, b, c) where $1 \leq a, b, c \leq 6$ and $a + b + c = 7$.

These are:

- (1,1,5), (1,2,4), (1,3,3), (1,4,2), (1,5,1),
- (2,1,4), (2,2,3), (2,3,2), (2,4,1),
- (3,1,3), (3,2,2), (3,3,1),
- (4,1,2), (4,2,1),
- (5,1,1).

There are 15 such triples.

Thus, the probability is $\frac{15}{216} = \frac{5}{72}$.

